

Reference Examples from the Exploration Tests

Content of this document

- Pages 2–7 is Local example pages - six selected phage bacterium pairs.

For each selected pair, the document shows explanation outputs obtained with different baselines and explainer settings. The purpose is to compare how stable the feature attributions remain across alternative configurations, rather than to draw final biological conclusions from a single plot

These pairs were chosen because they appear both in the randomly generated test dataset used during ensemble model training and in the related XAI study prepared by Federico, which makes qualitative cross-checking possible.

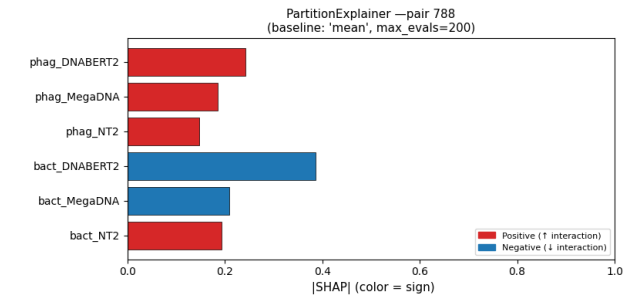
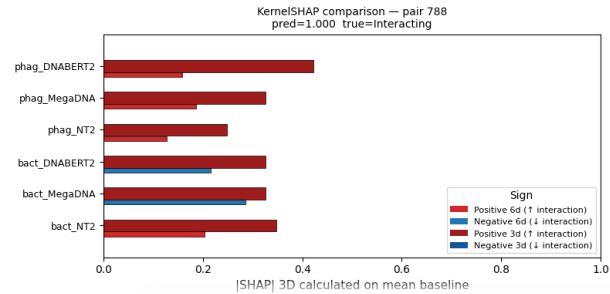
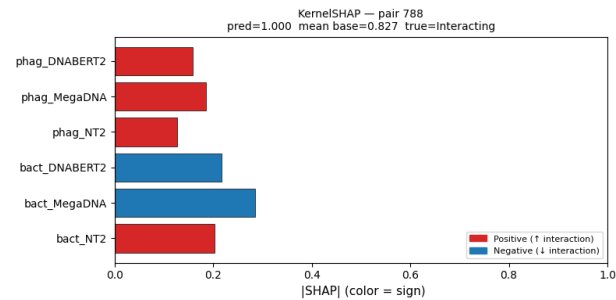
- Pages 8–9 present Global explanations summarized at dataset level.

The last two pages show global views organised by confusion-matrix sections and aggregated attribution summaries. One page uses the mean-ablation setting with the zero baseline (Nb1), while the other uses the hierarchy-SHAP setting with a balanced random baseline of 500 samples, split equally between positive and negative interactions (NB2).

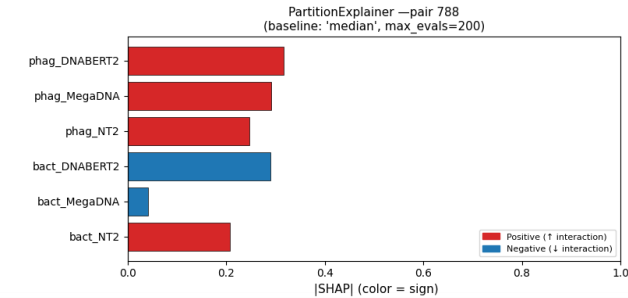
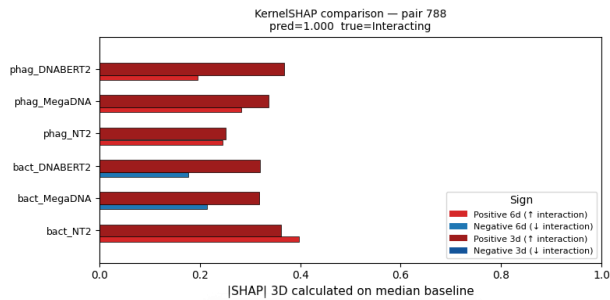
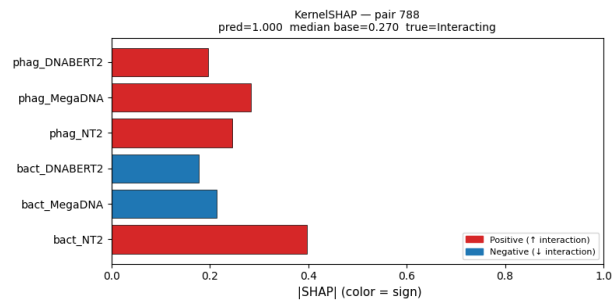
Reading Note: The zero baseline was retained because it gave the most stable behaviour in some tests, although it makes the true-negative section not representative. Detailed interpretation and the main analytical conclusions are provided in the report; this PDF is primarily intended for inspection, comparison, and documentation of the generated examples.

Pair[0] id788 = b153/p2014

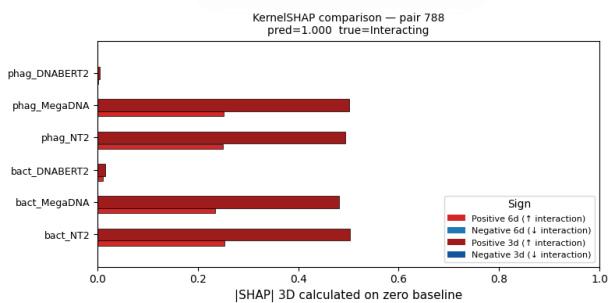
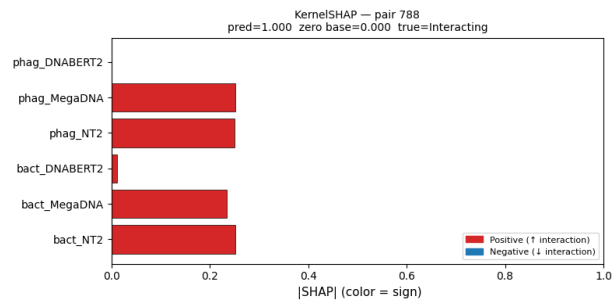
Per-dimension mean baseline



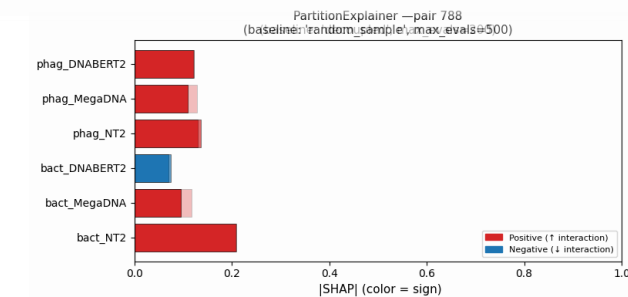
Per-dimension median baseline



Zero baseline



Hierarchy-SHAP (balanced 500 over decoupled)

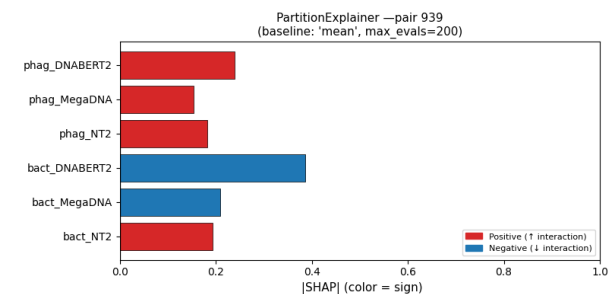
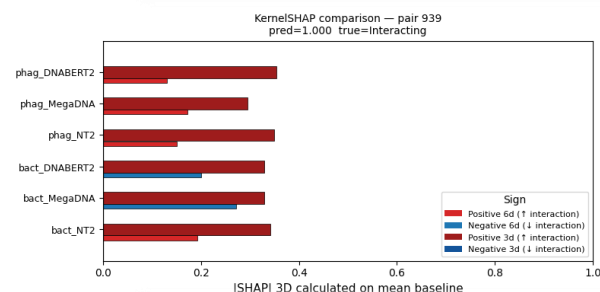
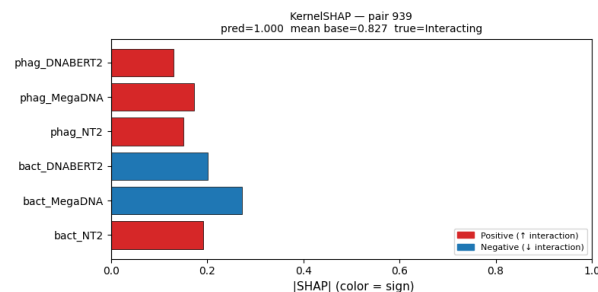


Mean ablation: both genomes (left) & one fixed (right)

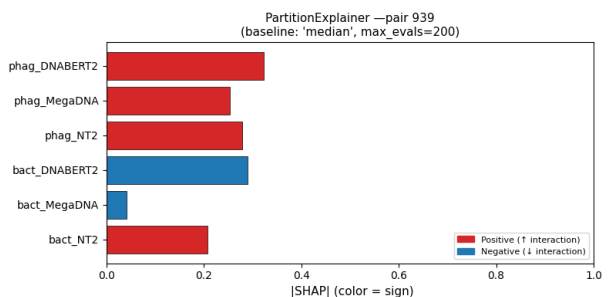
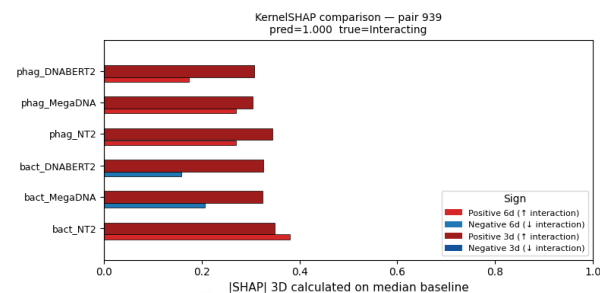
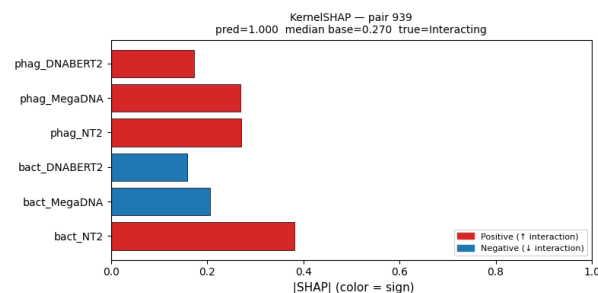
Hierarchy-SHAP

Pair[1] id939 = b153/p2153

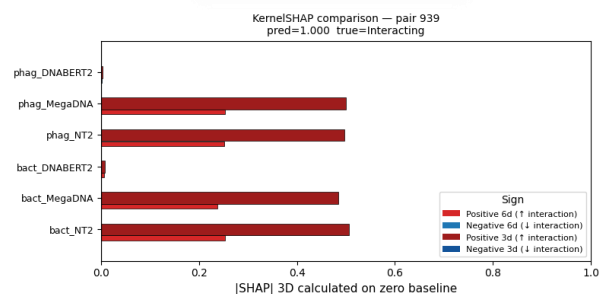
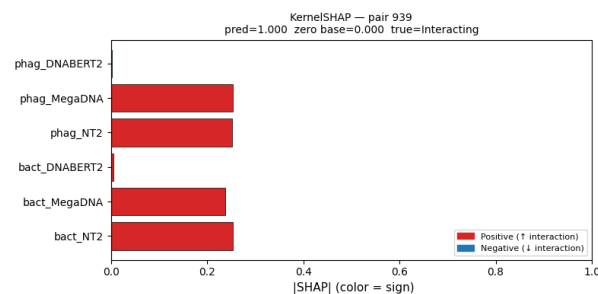
Per-dimension mean baseline



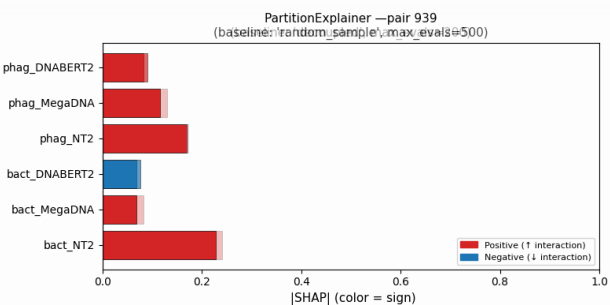
Per-dimension median baseline



Zero baseline



Hierarchy-SHAP (balanced 500 over decoupled)

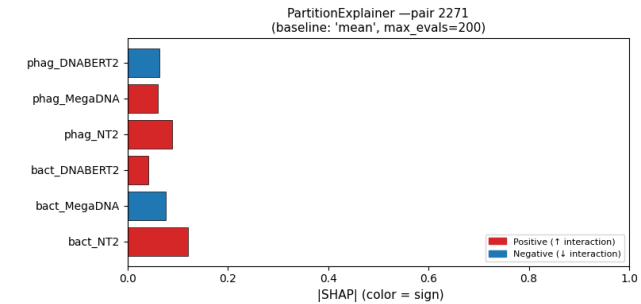
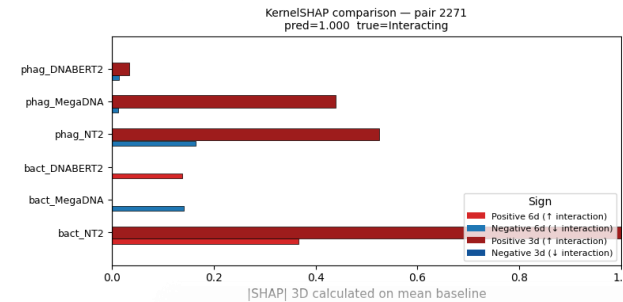
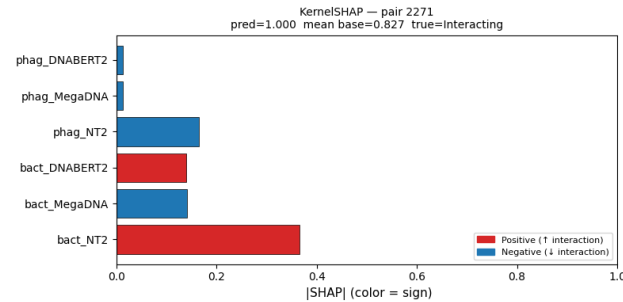


Mean ablation: both genomes (left) & one fixed (right)

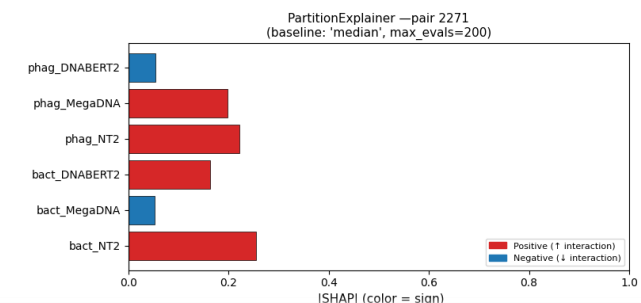
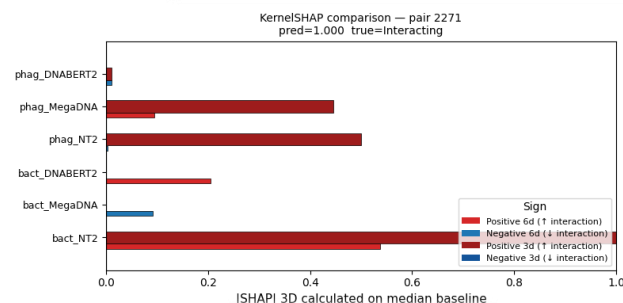
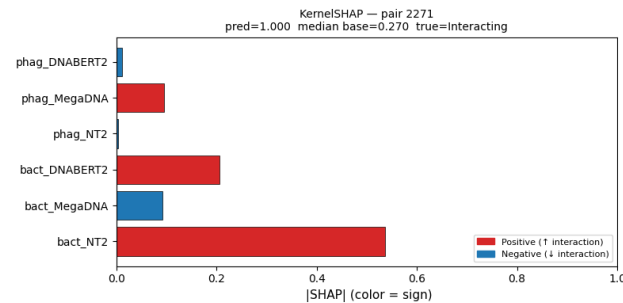
Hierarchy-SHAP

Pair[2] id2271 = b1869/p6187

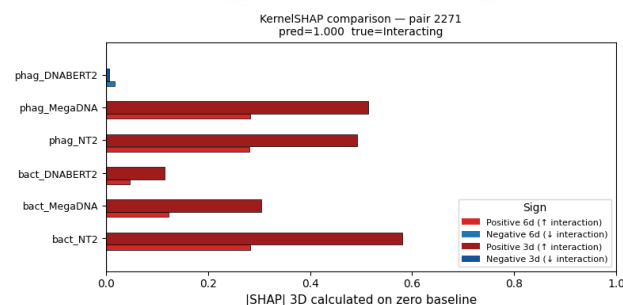
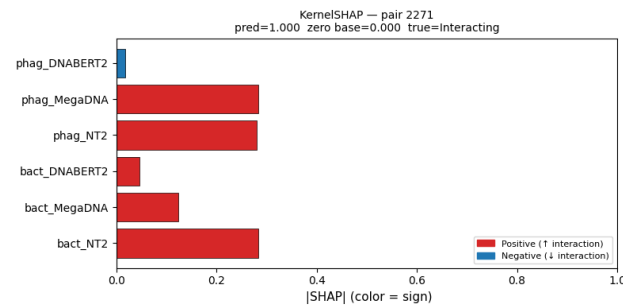
Per-dimension mean baseline



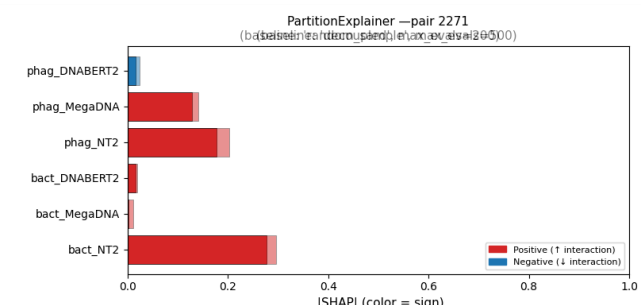
Per-dimension median baseline



Zero baseline



Hierarchy-SHAP (balanced 500 over decoupled)

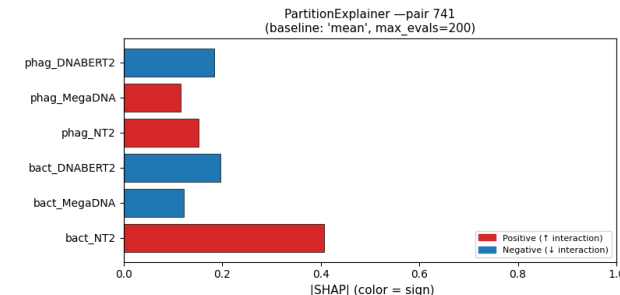
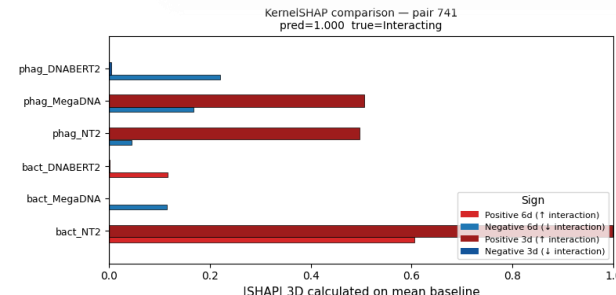
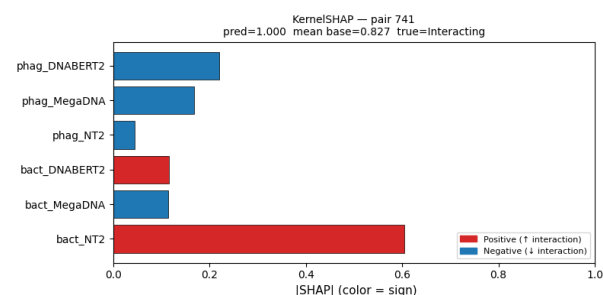


Mean ablation: both genomes (left) & one fixed (right)

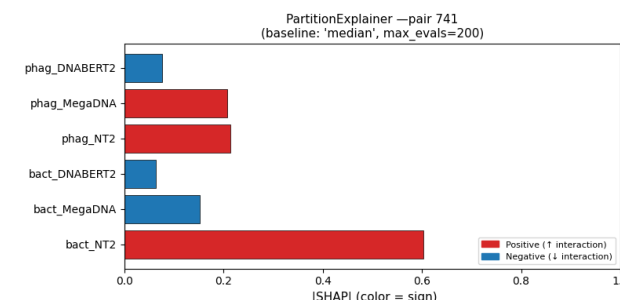
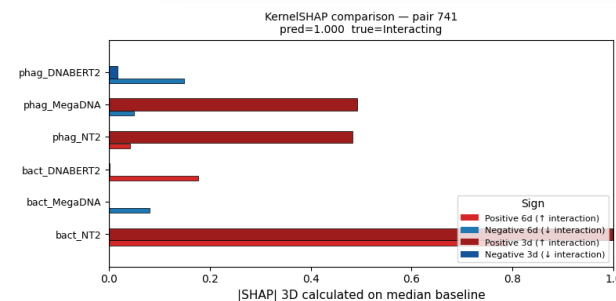
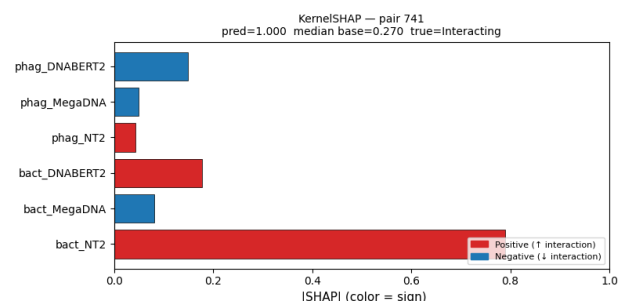
Hierarchy-SHAP

Pair[3] id741 = b5859/p451

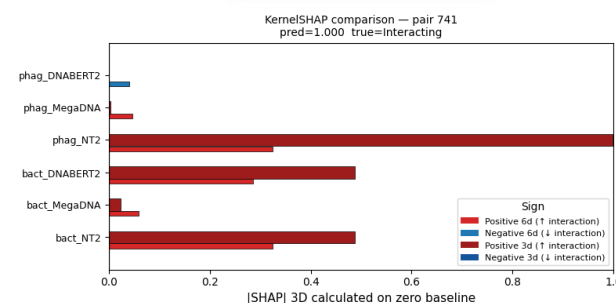
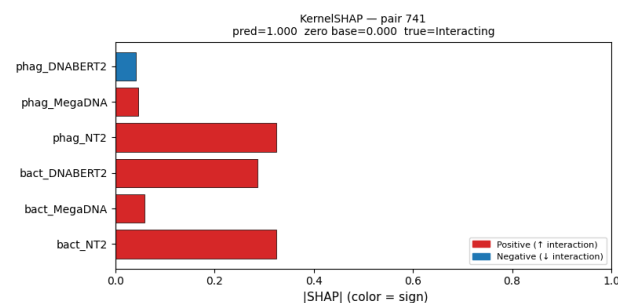
Per-dimension mean baseline



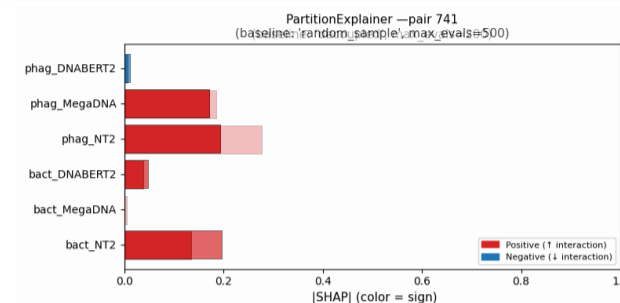
Per-dimension median baseline



Zero baseline



Hierarchy-SHAP (balanced 500 over decoupled)

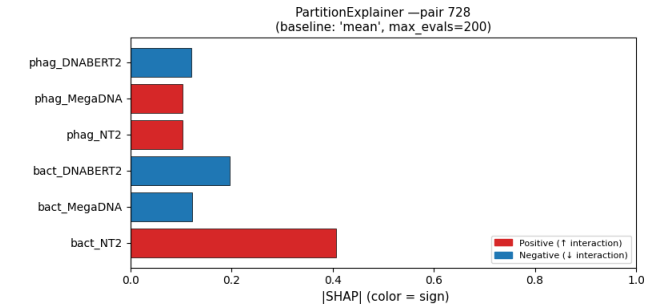
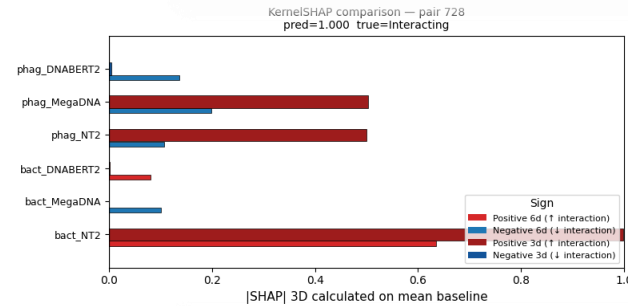
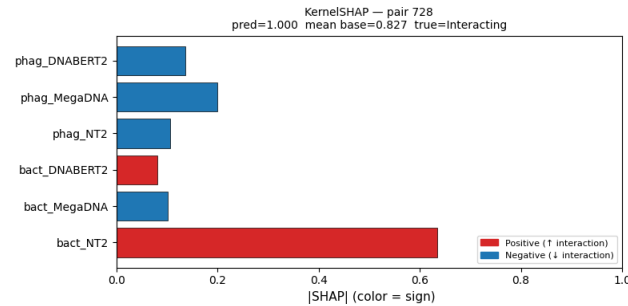


Mean ablation: both genomes (left) & one fixed (right)

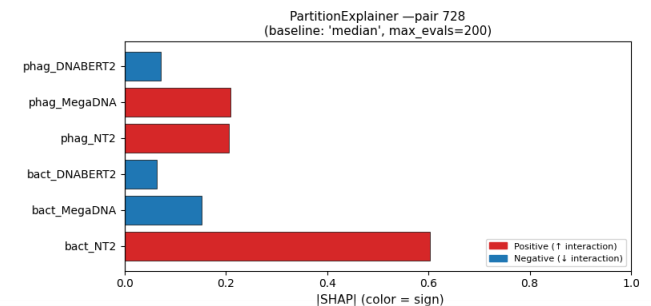
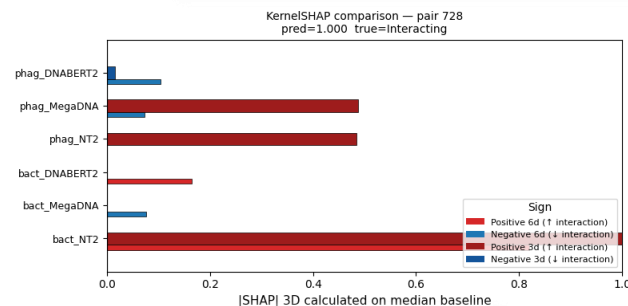
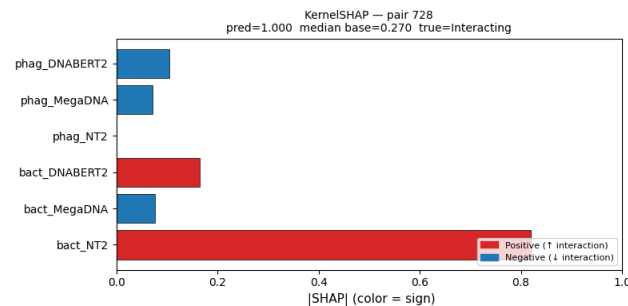
Hierarchy-SHAP

Pair[4] id728 = b5859/p5924

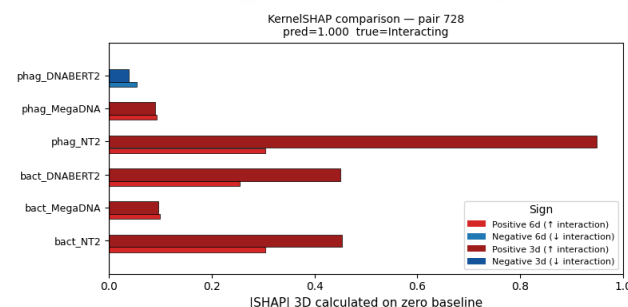
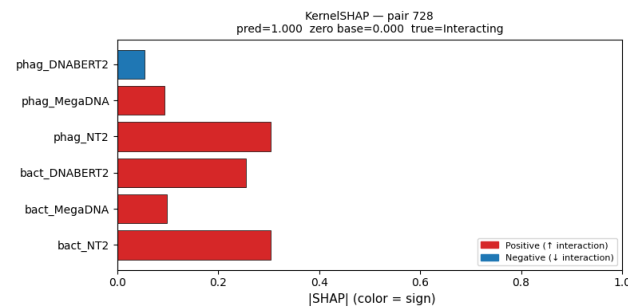
Per-dimension mean baseline



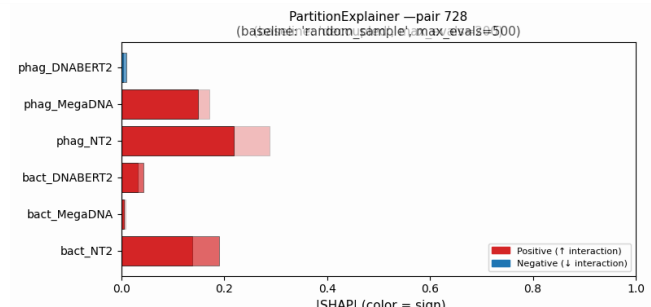
Per-dimension median baseline



Zero baseline



Hierarchy-SHAP (balanced 500 over decoupled)

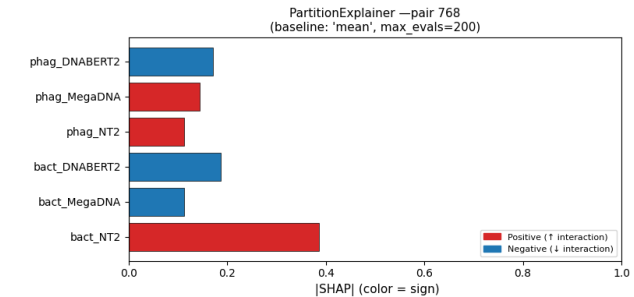
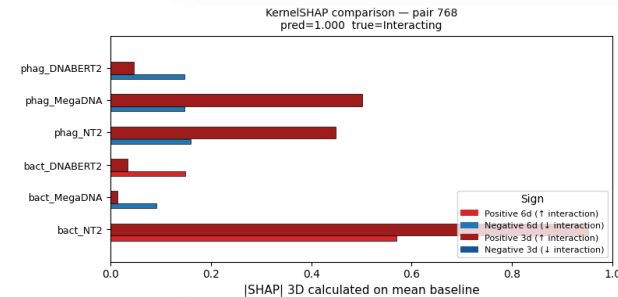
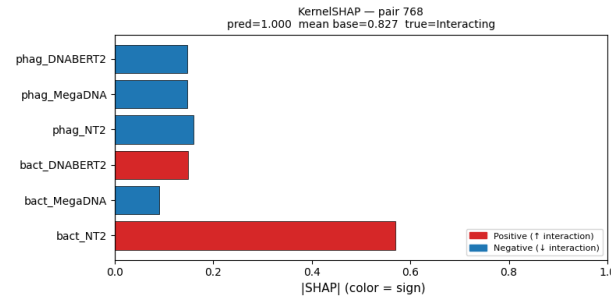


Mean ablation: both genomes (left) & one fixed (right)

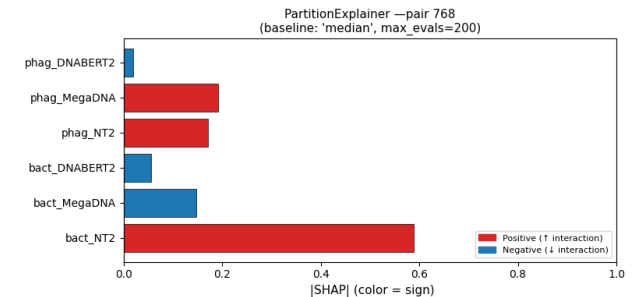
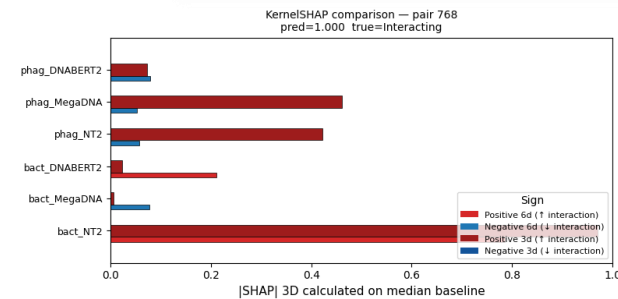
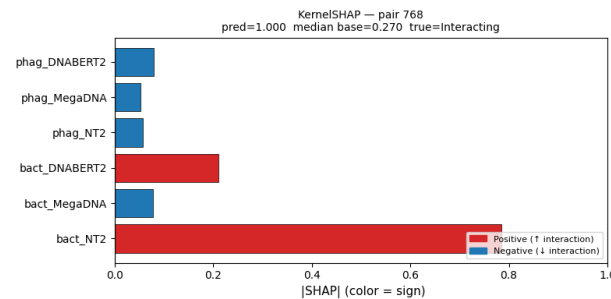
Hierarchy-SHAP

Pair[5] id768 = b5859/p5928

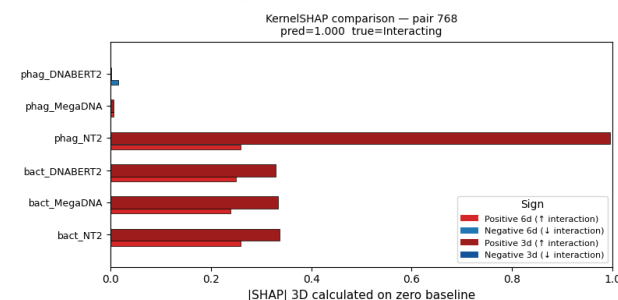
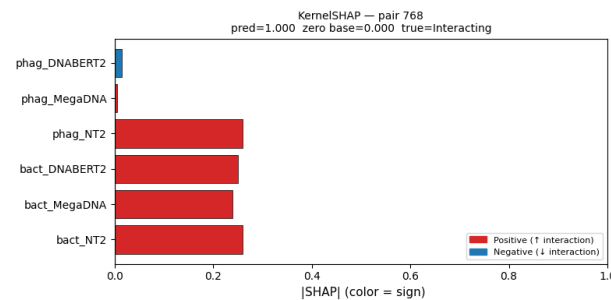
Per-dimension mean baseline



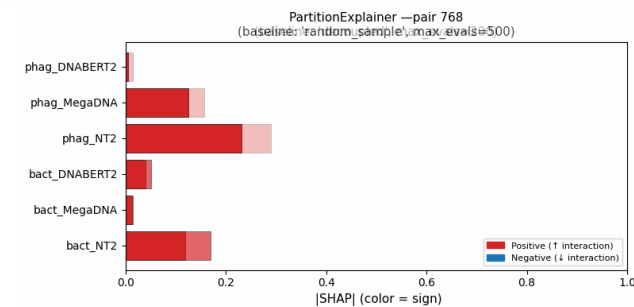
Per-dimension median baseline



Zero baseline

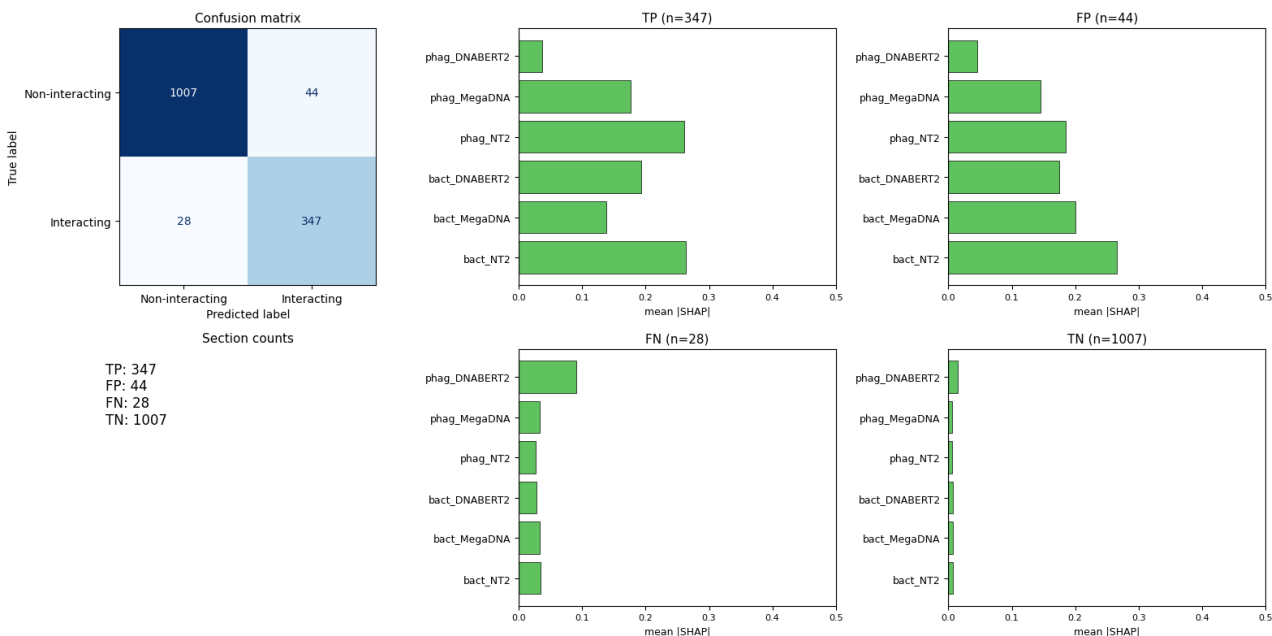
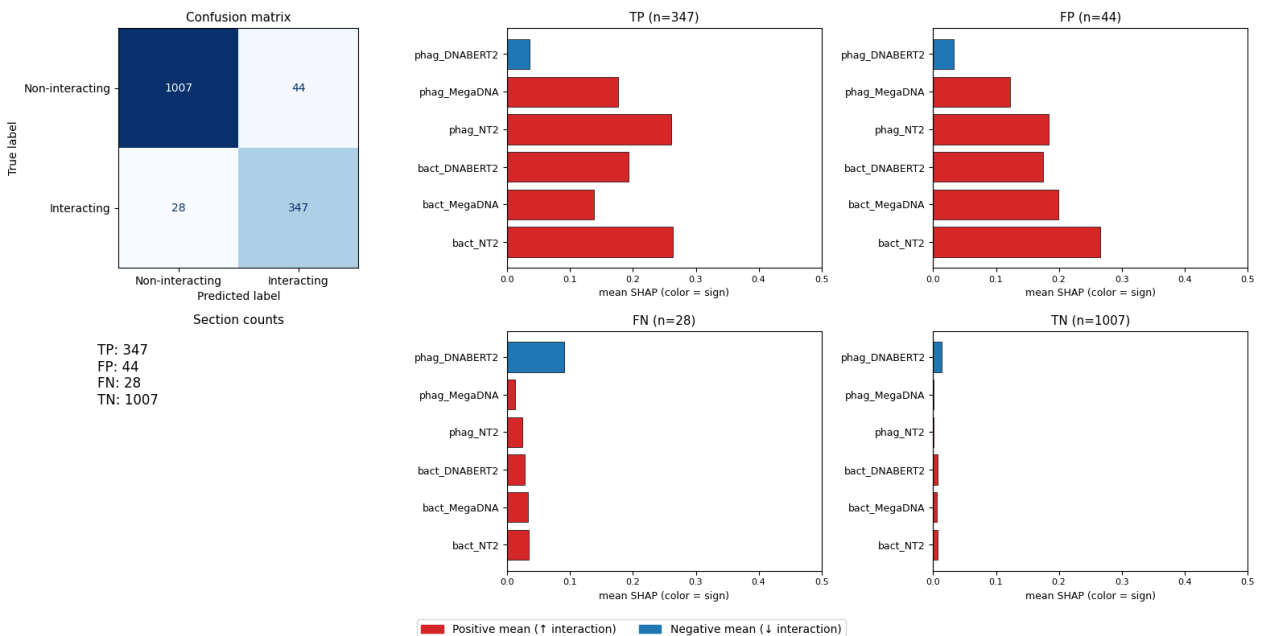


Hierarchy-SHAP (balanced 500 over decoupled)

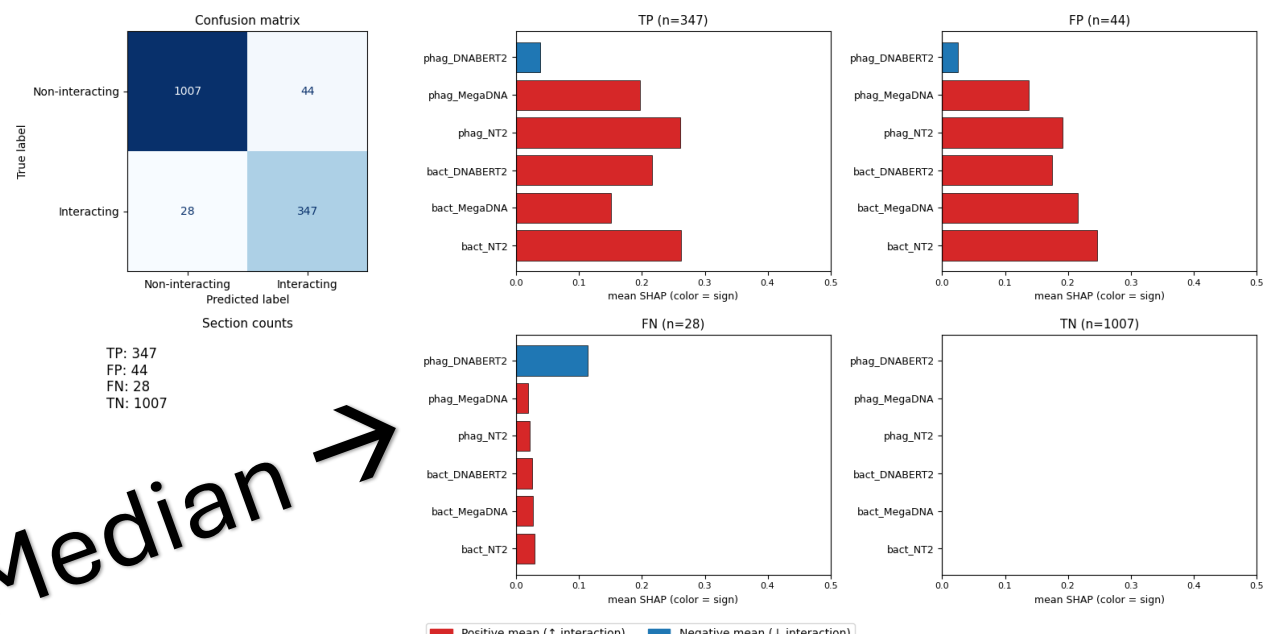
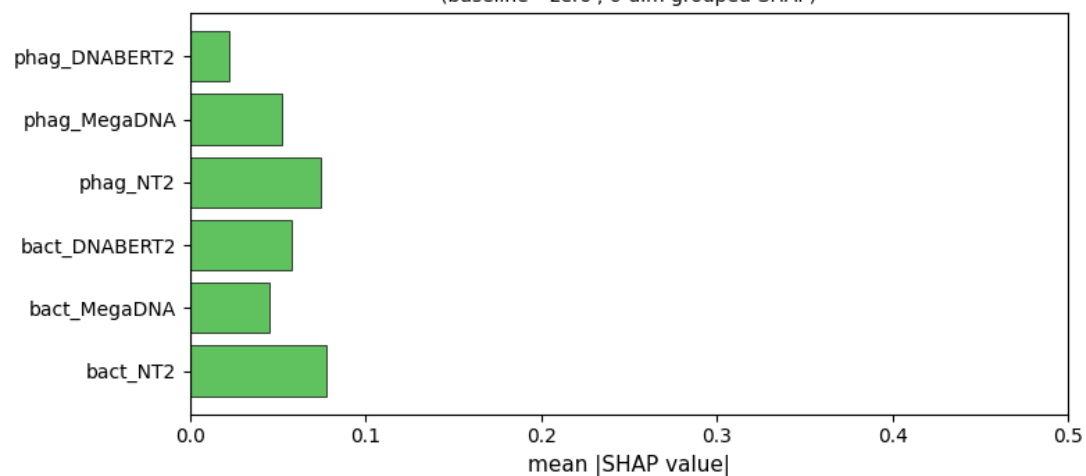


Mean ablation: both genomes (left) & one fixed (right)

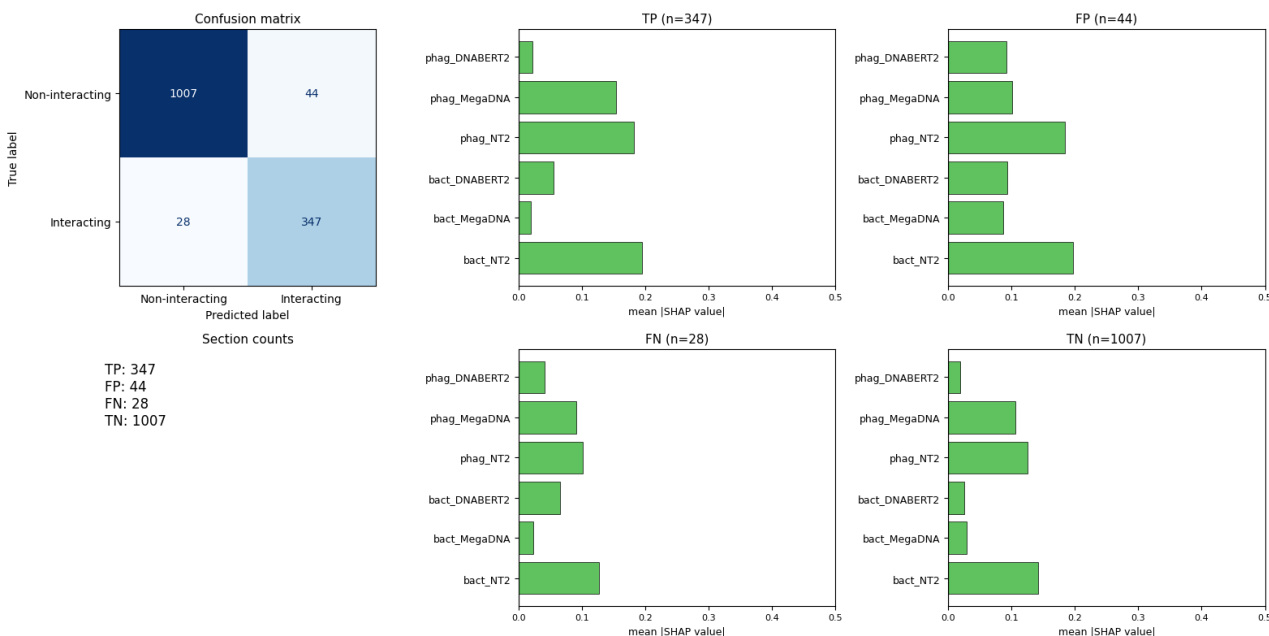
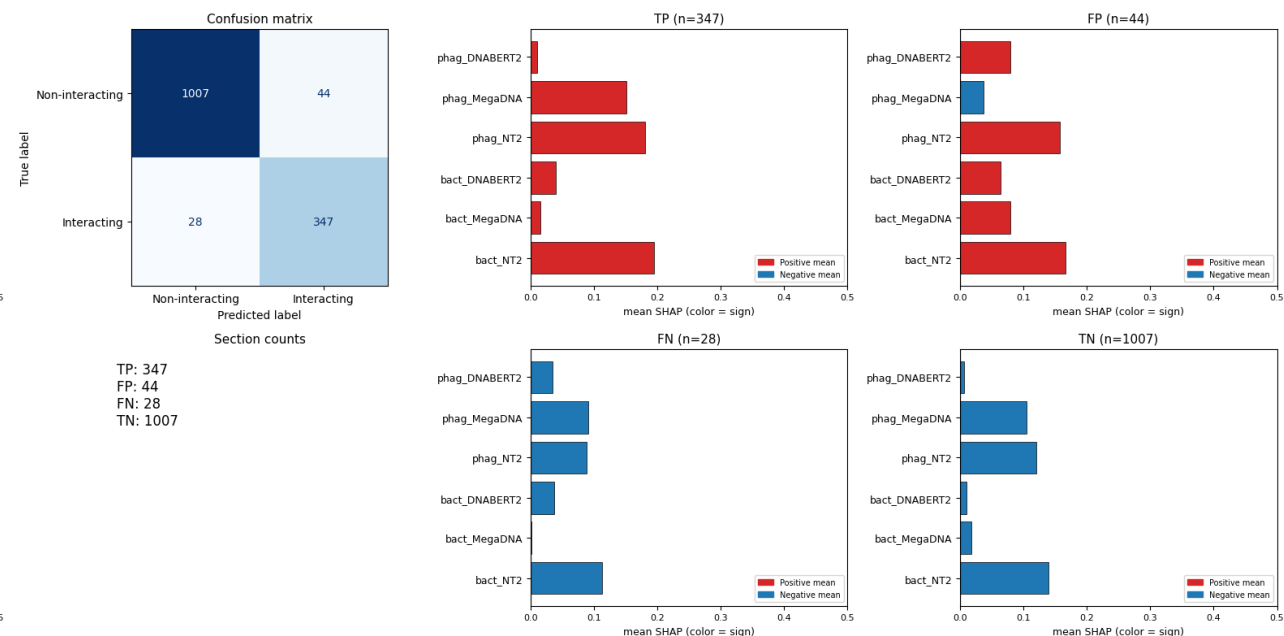
Hierarchy-SHAP

Global explanation by confusion-matrix section
(baseline='zero', threshold=0.5, 6-dim mean ablation)Global explanation by confusion-matrix section
(baseline='zero', threshold=0.5, 6-dim mean ablation)Global explanation by confusion-matrix section
(baseline='zero', threshold=0.5, 6-dim mean ablation)

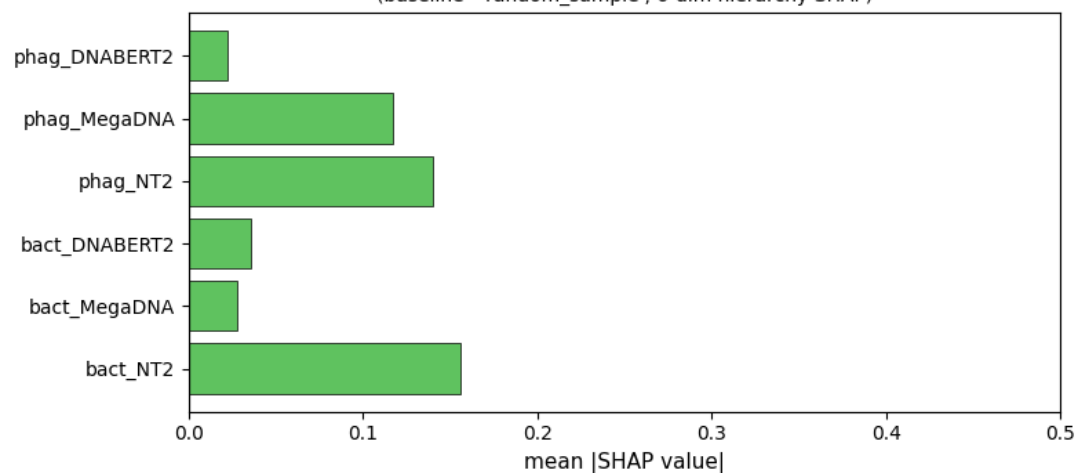
Results generated by Nb1

Global mean |Ablation| over all pairs
(baseline='zero', 6-dim grouped-SHAP)

Median →

Global explanation by confusion-matrix section
(baseline='random_sample', threshold=0.5, 6-dim hierarchy-SHAP)Global explanation by confusion-matrix section
(baseline='random_sample', threshold=0.5, 6-dim hierarchy-SHAP)Global explanation by confusion-matrix section
(baseline='random_sample', threshold=0.5, 6-dim hierarchy-SHAP)

Results generated by Nb2

Global mean [SHAP] over all pairs
(baseline='random_sample', 6-dim hierarchy-SHAP)

Median →

